

ContourMark: Blind, Representation-Invariant Watermarking of SVG Geometry

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Abstract

Language models and specialist generators now emit SVG code at scale, and regulation (EU AI Act Art. 50) asks providers to mark AI-generated content in a machine-readable way. Current practice for SVG is signed metadata, which the default configuration of the dominant optimizer (SVGO) deletes. Token-level LLM watermarks fail for the same reason: SVG toolchains rewrite nearly every token while preserving the picture. We present ContourMark, a blind watermark carried by the drawn geometry itself. Each contour is resampled by arc length. The magnitudes of a band of Fourier (or cosine) coefficients of its normalized radial function are invariant to command syntax, segment subdivision, path merging and reordering, start point, direction, and similarity transforms. A keyed dither-modulation quantizer moves these magnitudes onto a secret lattice. A representation-free capacity model lets the detector decide, from the observed geometry alone, which coefficients a contour could have carried. Detection needs only the SVG and the key. It reports a p-value that is a valid upper bound over the key’s randomness for any SVG chosen independently of the key, computed exactly by an exponentially tilted convolution rather than fitted to data. On 900 icons, logos, and emoji from nine open-license sets, 84% of marked assets are detected at $p \leq 10^{-6}$ from the SVG and key alone. The misses are mostly icons with at most one curved contour. Of the detectable assets, 94% survive default SVGO, 100% survive Scour, and 100% survive rotation, scaling, mirroring, reordering, subdivision, and path merging or splitting. Metadata, XML-comment, coordinate-LSB, Bézier-split, and vertex Fourier-descriptor baselines all fail under SVGO, rounding, or similarity transforms. Over 268,200 key-randomized null tests the empirical false-positive rate tracks the bound from 10^{-1} to 10^{-5} . On SVGs written by a local LLM, 24 of 40 files use clipping; detection is 78% when clipped contours are excluded and 93% when a renderer resolves their visibility. We report the failure boundary as prominently as the successes: noise at the mark’s amplitude, aspect-ratio stretching, stroke outlining, rasterize-and-retrace, an informed re-quantization attacker, and icons made only of straight lines.

1 Introduction

SVG is both a program and a picture. The same rendering admits absolute or relative commands, shorthand curves, arbitrary numeric precision, merged or split paths, group transforms, and basic shapes instead of paths. Production pipelines exercise all of this freedom. SVGO, the default optimizer in web build chains, rounds coordinates, rewrites commands, merges paths, and strips metadata and comments. As a result, a provenance signal stored in syntax (a C2PA manifest, an XML comment, least-significant digits, a segment-split pattern) disappears on the first build.

The problem is becoming practical. Code-writing LLMs routinely produce SVG. Specialist models (StarVector, OmniSVG, InternSVG) generate icons and illustrations. Article 50 of the EU AI Act, applicable from August 2026, requires machine-readable marking of AI-generated content. Text watermarks that bias token sampling do not transfer: an SVG optimizer is a near-total semantics-preserving rewrite of the token stream, the extreme case of the code-rewriting attacks that already defeat code watermarks.

ContourMark moves the mark from tokens to *drawn geometry*. It is applied post hoc to a generator’s output, so it needs no model access. It applies only to path geometry with enough curved contours to carry evidence (§5); text, raster images, and straight-line drawings are out of scope. Our contributions are:

1. **A representation-invariant carrier.** The band magnitudes of a contour’s arc-length radial function do not depend on how the contour is encoded (§3.1).
2. **Keyed QIM with content-seeded dithers and a representation-free capacity model.** The verifier recomputes, from geometry alone, which coefficients a contour can carry. A structure-preserving embedder realizes the mark on the file’s own control points (§3.2).
3. **An exactly calibrated blind detector.** It returns a valid p-value bound over key randomness, computed by tilted discretized convolution, and combines a dense test with a Bonferroni per-contour test for composed documents (§3.3).

4. **A corpus-scale evaluation** on 900 real assets and LLM-generated SVG. It covers 36 keyless transformations, four re-implemented prior schemes, fidelity and size costs, a 268,200-test key-randomized null calibration, and an informed attacker (§5).

Two inference-time variants, with their own and much thinner evidence, are described in §4.

2 Threat model and goals

A provider holds a secret key K . It marks SVGs before release, and later tests a suspect file with K alone (no original, database, or manifest). We distinguish three adversaries:

- **A0 (benign processing):** optimizers (SVGO, Scour, pcosvg), precision reduction, format normalization, similarity transforms, reordering, merging and splitting, subdivision, composition into larger artwork, cropping.
- **A1 (keyless editing):** coordinate noise, simplification to polylines, non-uniform scaling, rasterize-and-retrace.
- **A2 (informed, keyless):** knows the algorithm and public parameters but not K ; for example, re-embeds with a random key.

We claim robustness only for named transformations and report A1/A2 failure boundaries. Following the impossibility results of Zhang et al., we make no claim against attackers with a quality oracle and unbounded edits.

False attribution is the primary security requirement. The probability that an SVG chosen independently of K is declared marked must be at most the stated α , without assumptions about the SVG distribution. Keys and asset identifiers should be committed before release if detection is used as evidence. **Visible-content binding.** The detector scores only contours whose visibility a stated SVG subset determines: inline paint, `display`, `visibility`, `opacity`, and the root viewport. Contours under `clip-path`, `mask`, `filter`, or a nested viewport are excluded, and documents with CSS rules, scripts, or animation return *indeterminate*, not a detection. An optional mode asks a renderer (librsvg) whether each such contour contributes pixels. Occlusion by later opaque shapes is not resolved in the strict mode. A positive result means that visible geometry marked with K is present; it does not attribute the whole document.

3 Design

3.1 Carrier: radial-function spectra

For each visible contour c (after flattening transforms and converting shapes), we sample $N = 256$ points p_n uniformly by arc length. We then form the normalized radial

function

$$\rho_n = \frac{|p_n - \bar{p}|}{\frac{1}{N} \sum_m |p_m - \bar{p}|}, \quad \bar{p} = \frac{1}{N} \sum_n p_n.$$

For closed contours the carrier is $m_k = 2|\hat{\rho}_k|$, where $\hat{\rho}$ is the DFT, for $k = 2, \dots, 9$. For open contours it is the magnitude of DCT-II coefficients.

Proposition 1. *The carrier is invariant, up to resampling error, to: path command syntax and precision-preserving rewrites; exact subdivision of segments; path merging, splitting and reordering; the start point and direction of closed contours; the direction of open contours; translation, uniform scaling, rotation and reflection.*

Proof sketch. Arc-length samples depend only on the drawn curve. Translation and scale are removed by centering and normalization, and rotation and reflection leave $|p - \bar{p}|$ unchanged. A change of start point is a cyclic shift and a reversal is an index reversal; DFT magnitudes are invariant to both, and DCT-II magnitudes are invariant to reversal. $\square$

Curves are sampled with a per-segment budget proportional to length, so the chord error is nearly independent of how the curve is split. Without this, arc-versus-cubic encodings differed by 10^{-3} in ρ , a quarter of the lattice step.

3.2 Keyed lattice and content-seeded dithers

Each contour receives a seed $s(c)$: a coarse, similarity-invariant shape class (quantized log isoperimetric or straightness ratio and log elongation). The dithers are $d_{s,k} = \text{PRF}_K(s, k) \in [0, 1)$. The embedder moves each carried magnitude to the nearest point of $\Delta(\mathbb{Z} + d_{s,k})$, with $\Delta = 0.004$ (in units of the mean radius). Seeding by content means an attacker cannot pool unrelated contours to estimate one lattice. Seed collisions within a document are handled exactly by the detector (§3.3).

Capacity from geometry. A circle drawn with four cubics cannot carry high-frequency ripple on its handles, and a polygon’s straight edges must stay straight. We define a *geometric deformation model* on the samples. Curved stretches move along their normals under a low-frequency field; exactly straight runs interpolate their endpoint motion, so lines stay lines. Coefficient k is carried if its magnitude Jacobian row, orthogonalized against lower frequencies, has norm ≥ 0.4 . The rule depends only on geometry, so the verifier recomputes it after any rewrite. The embedder marks the superset with gain ≥ 0.25 (hysteresis).

Structure-preserving realization. The embedder moves the file’s own handles by a low-frequency normal field. It solves for the field with minimum-norm Gauss-Newton on the complex coefficients. Where a magnitude is zero it linearizes along the direction of largest gain;

every band coefficient of a circle is such a case. Curved segments are split, exactly, only when the handles cannot realize a selected coefficient. Arcs are converted to $\leq 45^\circ$ cubics, because moving an arc’s endpoints keeps it rigid. Output precision is chosen per element so that rounding stays below $0.05 \Delta \bar{r}$. Lines stay lines, other elements and attributes are untouched, and comments are preserved.

3.3 Detection and exact calibration

Contours sharing a seed share dithers, so they are pooled. For group g and coefficient k , let

$$A_{g,k} = \sum_{c \in g} \mathbb{I}[k \text{ selected for } c] e^{2\pi i m_{c,k}/\Delta},$$

$$T = \sum_{g,k} \text{Re}(A_{g,k} e^{-2\pi i d_{g,k}}).$$

Theorem 1. *Let an SVG be fixed independently of K , and model the PRF as random. Then $T \stackrel{d}{=} \sum_{g,k} |A_{g,k}| \cos(2\pi U_{g,k})$ with $U_{g,k}$ i.i.d. uniform. Hence any upper bound $\bar{P}(t) \geq \Pr[\sum w_i \cos 2\pi U_i \geq t]$ with $w = |A|$ yields a p -value satisfying $\Pr[\bar{P}(T) \leq \alpha] \leq \alpha$.*

Proof. Given the document, the $A_{g,k}$ are fixed. Distinct (g,k) have independent uniform dithers, and a uniform phase shift of a fixed phasor gives a uniform angle. $\square$

We compute $\bar{P}$ by convolution rather than asymptotically. Each term $w \cos(2\pi U)$ is rounded up to a grid. The discretized sum then stochastically dominates T , so its tail is a valid bound. We exponentially tilt each term by the Chernoff-optimal θ , convolve by FFT (so floating point is accurate near t), and untilt. The reported bound is the minimum of this value and the Chernoff bound. Against 2×10^6 Monte Carlo draws it is within 10% of the truth and never below it; Chernoff alone is about $5\times$ conservative. The implementation uses double-precision FFTs and rounds weights to 12 decimals. We have not produced a formal floating-point error bound, so “exact” here means exact up to that arithmetic; the Chernoff bound is the rigorous fallback.

Composition. A marked icon pasted among unmarked artwork dilutes T . We therefore also compute a per-contour bound p_c and use $p = 2 \min(\bar{P}(T), C \min_c p_c)$, which is valid by the union bound under any dependence. This raised the detection of a composed stroke icon from 10^{-2} to 10^{-10} .

4 Inference-time variants

The spectral method above changes a finished SVG. Two further mechanisms act while a generator samples; both move no geometry after generation. They are separate systems with separate evidence, and the 900-asset results in §5 do not validate either.

Candidate-manifest sampler. When a generator proposes m candidate contours with probabilities

q_i , a keyed Gumbel-max draw $J = \arg \max_i \log q_i - \log(-\log u_i)$ picks one. Verification needs a private authenticated candidate manifest and uses a Poisson-binomial test. In three 16-step Qwen-generated drawings all keyed choices verified after SVGO, Scour, rounding, reordering, and bounded translation.

Blind geometric sampler (geosample). For point-token generators (IconShop, OmniSVG), each candidate endpoint token completes a vertex, and each candidate first control point fixes a tangent at that vertex. Candidates are grouped by a quantized descriptor: interior angle, $|\log|$ arm ratio and summed bulge for a vertex; angle and $|\log|$ ratio of end derivatives for a tangent. A group is chosen by keyed Gumbel-max on group mass and a token inside it by ordinary sampling. Detection needs only the SVG and K : the sum of $-\log(1 - u_b)$ over *distinct* recovered descriptors is $\text{Gamma}(n, 1)$ under the null.

Distribution. Reusing a descriptor’s keyed uniform at every step preserves each step in isolation but not the sequence: in a two-step example the second-step probability is 0.5 where ordinary sampling gives 0.7. We therefore use a keyed uniform only the first time its descriptor is examined within a drawing, and a fresh unkeyed uniform afterwards. A uniform at a never-examined index is independent of the generated prefix, so by induction the joint output law equals the model’s, in the random-PRF idealization. This costs power: on a toy point-token generator with about 100 distinct descriptors the median clean $\log_{10} p$ is -18.7 , against -49 with naive reuse.

Evidence so far. On that toy generator (30 drawings), detection at 10^{-6} is 100% clean and after SVGO, Scour, picosvg, rounding to one decimal, rotation, scaling, mirroring, reversal, reordering, merging and splitting. It is 27% under handle noise at 0.1% of the diagonal, and 0% under exact midpoint subdivision, a 1.2:1 stretch, polyline flattening, and retracing. The descriptors are invariant to similarity transforms, reversal, and degree elevation; they are not intrinsic to the curve under subdivision. A key-randomized null on the 900 human icons is reported in §5. *Real model.* We ran the public IconShop checkpoint (30 prompts, 4 seeds, 120 plain and 120 marked icons per setting). At the model’s default top- p of 0.5, 11% of marked icons are detected at 10^{-6} and 47% at 10^{-3} . At top- p 0.9, 62% and 91% are, and 85% of drawings with at least 80 distinct descriptors. Detection is unchanged after SVGO (all four settings), Scour, picosvg, rounding to any precision down to integers, reordering, reversal, merging and splitting. It is lost under subdivision, polyline flattening, retracing, a non-uniform stretch, and handle noise at 0.1% of the diagonal. Composition with three unmarked icons keeps 65% of the clean detections at top- p 0.9. CLIP prompt-retrieval accuracy is the same for marked and plain icons (44% against 45% at top- p 0.5, 40% against 40% at 0.9), consistent with the distribution-preservation theorem; the intervals are about ± 10 points at this sample size. A key-randomized null over the plain samples (36,000 tests per setting) gives 0.8% at 10^{-2} and 0.08–

0.10% at 10^{-3} . Power on low-entropy decoding, robustness to subdivision, and adaptive removal remain open for this sampler.

5 Evaluation

5.1 Setup

Corpus. 900 test assets, 100 from each of nine open-license sets: Lucide and Tabler (stroked line icons); Bootstrap, Material, Font Awesome, Remix, and Octicons (filled icons); Simple Icons (logos); and OpenMoji (colour illustrations). A disjoint 270-asset development split, separated by filename hash, was used for all tuning. Parameters were frozen before any test-split run. We also use 40 SVGs written by a local LLM (Qwen 3.5 4B through Ollama) from icon prompts.

Methods. We compare ContourMark (defaults $\Delta=0.004$, $k=2..9$) with: an XML comment; a `<metadata>` element as a placeholder for metadata-borne credentials (it is not a C2PA implementation); geometry-aware coordinate-parity LSB (64 chips at 10^{-3}); Bézier-split steganography; and our adaptation of a vertex Fourier-descriptor magnitude watermark with correlation detection (after Solachidis/Doncel/Pitas; not verified against the authors’ equations). The detection threshold for all methods is $p \leq 10^{-6}$. In the main table baselines run at fixed default strengths and are *not* distortion-matched to ContourMark: their median PSNR at 1024 px is 54–83 dB against 25 dB for ContourMark. That table shows which carriers survive which transformations; the matched-distortion comparison is reported separately below.

Attacks (36, all keyless):

- optimizers: SVGO (default, multipass, precision 2 and 1), Scour (default, precision 3), picosvg;
- absolute rounding to 2, 1, and 0 decimals;
- similarity: translate, scale $\times 0.37$ and $\times 3$, rotate 30° and 90° , mirror, nested group transform;
- non-uniform 1.2:1 stretch;
- structure: reorder, reverse, restart, exact subdivision, merge, split;
- deletion of 25% or 50% of contours, half crop, composition into a 2×2 grid with unmarked icons;
- Gaussian handle noise at 0.1, 0.3, and 1% of the diagonal;
- flattening to polylines with Douglas–Peucker simplification;
- rasterize (librsvg, 512 or 1024 px) and retrace (potrace).

5.2 Detection and robustness

ContourMark detects 84% of marked test assets (755/900) with no attack. Across sources this ranges from 70% (Lucide) and 71% (Tabler), which are mostly straight strokes, to 93% (OpenMoji). Among the 145 misses:

- 57 have at most one markable contour;
- the median miss still reaches $p = 4.8 \times 10^{-4}$;
- 83 reach $p \leq 10^{-3}$.

Capacity, not robustness, limits coverage. The embedder reports the attainable p in advance, so a provider knows when an asset cannot be marked.

Conditional on clean detection, survival rates with 95% Wilson intervals are:

- SVGO default 93.8% [91.8, 95.3]; SVGO multipass 93.9%;
- Scour 100% [99.5, 100]; picosvg 78.5%;
- 2-decimal rounding 75.8%;
- all seven similarity transforms 100%;
- reorder, reverse, subdivide, merge, split 100%; restart 96.2%;
- deleting 25% of contours 96.8%; removing contours in one half 82.9%; composition 89.1%;
- light polyline flattening (0.05% tolerance) 82.3%.

Figure 1 compares attack families.

The failure boundary is equally clear:

- 1-decimal rounding: 14.8%;
- SVGO precision 1: 8.7%;
- handle noise at 0.1% of the diagonal: 0.5%;
- the 1.2:1 stretch: 0%, since the carrier is only similarity-invariant;
- rasterize-and-retrace: 10.6% at 512 px and 23.6% at 1024 px.

Baselines. Both metadata-style marks are removed by default SVGO and picosvg. Coordinate LSB detects 80% clean and 54% after SVGO, which leaves the chosen coordinates on its 10^{-3} grid. It detects 0% after 2-decimal rounding, rotation, down-scaling, subdivision, or composition. It partly survives transformations that keep coordinates on its grid: 70% after $\times 3$ scaling and mirroring, 34% after translation, and 80% under an unflattened group transform. Bézier splitting detects only 9% clean at 10^{-6} (icons have few cubics) and 0% after SVGO. Our vertex Fourier-descriptor adaptation detects 0% even without attack at its default strength.

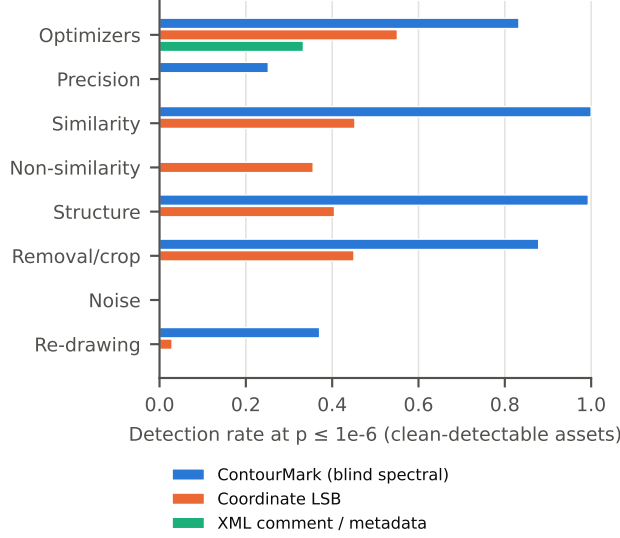


Figure 1: Detection at $p \leq 10^{-6}$ by attack family, conditional on clean detection for each method (900 test assets).

Matched distortion. Because those defaults differ in how far they move the drawing, we swept every method’s strength on 54 development icons and measured the same quantity for all: the 95th-percentile symmetric boundary distance, as a fraction of the diagonal.

- At about 0.16%, ContourMark ($\Delta=0.004$) detects 91% clean, 85% after SVGO, 70% after 2-decimal rounding, and 91% after rotation and subdivision.
- Coordinate LSB at its closest setting (0.03%, quantum 10^{-2}) detects 78%, 69%, 69%, and 0% after rotation or subdivision. Raising it to 0.35% does not change the last two.
- The vertex Fourier-descriptor adaptation detects 0% on icons at every strength up to 1.3% distortion. On 600-vertex broadband polylines, the regime it was designed for, it detects 33% at its highest strength, and those detections survive rotation. It is a working method without an operating point at icon scale.
- An ablation that keeps our arc-length carrier but replaces QIM with additive spread spectrum detects 0% at every strength, on icons and on long polylines. Host interference, which QIM removes, is what defeats correlation detectors here.

These sweeps use the development split and four transformations; they are not yet a test-split result.

5.3 Calibration

For each unmarked test document we computed the geometric observations once and scored them under 300 in-

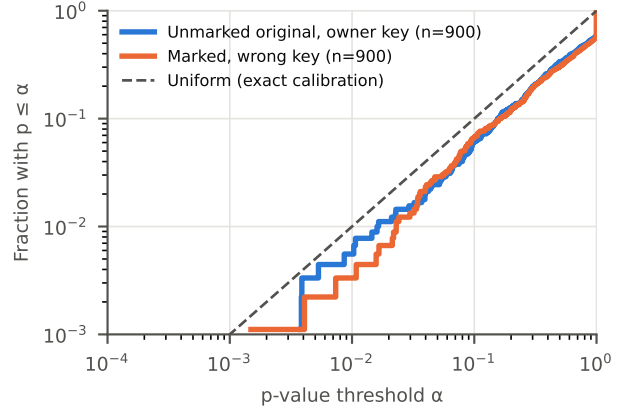


Figure 2: Empirical null CDF of reported p-values (log-log). Points on or below the diagonal indicate a valid, conservative bound.

dependent keys, giving 268,200 null tests over 894 documents with at least one carrier. Observed rates were:

- 6.9% at $\alpha=10^{-1}$;
- 0.68% at 10^{-2} ;
- 0.063% at 10^{-3} ;
- 4.5×10^{-5} at 10^{-4} ;
- 1.1×10^{-5} (3 hits) at 10^{-5} .

The bound holds or is met within sampling noise at every level. The smallest null p-value was 9.3×10^{-7} . Under exact calibration the expected count at 10^{-6} is at most 0.27, so observing one has probability about 0.24. The per-asset nulls in the main run agree (Figure 2): the unmarked original with the owner key gives 0.6% at 10^{-2} and none at 10^{-6} ; the marked file with a wrong key gives none at 10^{-6} .

5.4 Fidelity, size, and cost

Distortion.

- The median maximum curve displacement is 0.30% of the drawing diagonal (95th percentile 0.68%): about 0.3px when a full-canvas icon is rendered at 64px, 1.1px at 256px, and 4.4px at 1024px.
- Median SSIM is 0.985 at 256px and 0.979 at 1024px; PSNR is 28.9 and 25.3 dB.
- PSNR is dominated by sub-pixel edge shifts on high-contrast line art. Side-by-side renders show no visible change at typical icon sizes.
- A perceptual study remains necessary before making an imperceptibility claim for large renders.

Size and time.

- Marked files are $1.9\times$ the original after SVG0, a median of 483 added bytes. The cost comes from curve subdivision and higher-precision coordinates.
- Joins that are smooth in the source stay smooth: the embedder constrains the two handles at such a join to a common tangent. Without this constraint marking introduced soft kinks ($0.5\text{--}12^\circ$) at up to half of the joins.
- Embedding takes 0.08s and detection 0.008s (medians, one core).

Operating curve. On the development split, the lattice step trades these costs directly:

Δ	Max shift (% diag)	PSNR @1024	SVG0	2 dp	0.1% noise
0.002	0.2	29.7	87%	53%	0%
0.004	0.3	26.0	92%	74%	0%
0.016	0.9	19.1	100%	88%	37%

Larger bands ($k=2..13$) and lower gain thresholds changed clean coverage by at most 1.1 points while increasing size, so the defaults use the smaller band.

5.5 LLM-written SVG

On 40 SVGs written by Qwen 3.5 4B, 24 use `clip-path`. Under the strict visibility rule, which neither marks nor scores clipped contours, ContourMark detects 78% clean, 78% after SVG0, and 62% after 2-decimal rounding. With renderer-assisted visibility for both embedding and detection it detects 93% clean, 93% after SVG0, and 78% after rounding. An earlier version of the detector scored clipped contours without checking that they are drawn and reported 100%; that number counted evidence that could be hidden, and we withdraw it. Byte overhead after SVG0 is larger than for icons because these files use basic shapes, which are rewritten as paths. Rasterize-and-retrace removes the mark from these overlapping colour drawings.

5.6 Informed attacker (A2)

An attacker who knows the algorithm and Δ but not K re-embeds with a random key. On 232 detectable assets (30 per source) this removes the mark from 98.3% of them (228). The attack’s distortion (median PSNR 26.4 dB at 1024 px) equals that of embedding (25.2 dB). With a doubled step the removal rate is 94.4% at 22.4 dB. Plain QIM with public Δ offers no margin against this attacker. Spread-transform dither modulation over keyed projections of the band is a candidate remedy that we have not implemented or measured. We scope our claims to A0 and A1 transformations.

5.7 Null for the geometric sampler’s detector

The `geosample` detector (§4) was scored on the same 900 unmarked human icons under 300 keys each (267,000 tests over 890 documents with descriptors, median 11 distinct descriptors). Observed rates were 6.3% at 10^{-1} , 0.68% at 10^{-2} , 0.072% at 10^{-3} , 7.9×10^{-5} at 10^{-4} , and 3.7×10^{-6} (one hit) at 10^{-5} . With this many trials the test cannot confirm rates near 10^{-6} empirically; that level rests on the Gamma null.

6 Limitations

- **Capacity.** Straight-line icons (a Lucide “minus”) carry nothing under a similarity-invariant carrier that keeps lines straight. Coverage is 69–92% by source. Relationships between contours (relative placement of strokes) are an unexploited carrier.
- **Similarity invariance only.** Non-uniform scaling and skew remove the mark. Per-contour affine normalization is possible but is unstable for thin contours.
- **Stroke outlining and retracing.** Converting strokes to filled outlines (picosvg, Illustrator “Outline Stroke”) keeps 78% of marks. Rasterize-and-retrace keeps at most 24%. A raster-domain decoder, as in FontCode and DeepMorph, would complement the geometric detector.
- **Informed removal.** Re-quantization with a random key removes the mark at embedding-level distortion (§5.6).
- **Seed classes.** Common shapes such as circles share seed classes across assets. An attacker holding many marked circles could estimate their lattice. Per-provider key rotation limits this; we have not measured it.
- **Evidence, not authorship.** A positive test shows that geometry marked with K is present, including after copy-and-paste into other work. Keys should be committed with timestamps if results are used as evidence.
- **Fidelity.** Distortion is measured with pixel metrics, and a perceptual study is still needed. Byte overhead ($1.9\times$ after SVG0) matters for icon sprites.

7 Related work

Vector graphics. Solachidis and Pitas, and Doncel, Nikolaidis and Pitas (TVCG 2007) watermark Fourier-descriptor magnitudes of *vertex sequences* with correlation detection. They are the closest prior art to our

invariants. Their carrier depends on the representation, and host interference requires hundreds of vertices; our re-implementation fails at icon scale. A 2024 V-system method represents Bézier curves exactly and reports transformation robustness only. GIS vector-map schemes quantize per-vertex angles or distance ratios with QIM and target simplification and cropping on dense maps. Blinova and Urbanovich hide data in cubic split ratios; this is lossless in rendering but fragile to re-parameterization and rounding.

Raster-decoded vector marks. FontCode, AutoStegaFont, GlyphShield, and DeepMorph learn perturbations of glyphs or drawings decoded from images, and target screenshots and print-scan. They need trained decoders and use empirical thresholds. They are complementary to our geometry-domain detector for the rasterize-and-retrace threat.

Quantization watermarking. Chen and Wornell’s QIM and dither modulation remove host interference. Spread-transform DM raises the cost of uninformed re-quantization. Content-derived keys (Fridrich) prevent pooling.

Generative-model marks. SynthID-Text, green-list, and Gumbel samplers mark tokens at inference; code watermarks fail under semantics-preserving rewrites. Image watermarks (Stable Signature, Tree-Ring, TrustMark, VINE) face regeneration attacks. Impossibility results bound what any scheme can claim.

8 Conclusion

Moving the mark from SVG syntax to drawn geometry turns the optimizer, the chief obstacle to SVG provenance, into a near no-op. A radial-function spectral carrier, keyed QIM, a representation-free capacity model, and an exactly calibrated detector give a post-hoc watermark that applies to any SVG generator. It is detectable from the file and key alone and survives the benign pipeline where every syntax-level baseline fails. The measured limits (capacity on straight-line icons, noise at the mark’s own amplitude, non-similarity warps, retracing, informed re-quantization) define the next steps: inter-contour carriers, spread-transform hardening, and a raster-domain decoder.